MIL-R-52003D(ME)
9 March, 1988
SUPERSEDING
MIL-R-52003C(ME)
22 May 1978

MILITARY SPECIFICATION

REELING MACHINE, CABLE, HAND

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a two-wheeled, hand-lift reeling machine for transporting reels of electrical cable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, and Seals.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-T-60	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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MILITARY

MIL-P-116	- Preservation Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-10924	- Grease, Automotive and Artillery.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-889	Dissimilar Metals.

2.1.2 Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ME

TA13217E7477	- Reeling Machine, Cable, Hand.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN-SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commerical Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed-to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between -the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The reeling machine shall be in accordance with Top Assembly TA13217E7477, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the reeling machine. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.1.2 Material deterioration prevention and control. The reeling machine shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the reeling machine may be exposed.

3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.1.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.1.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the reeling machine may be newly fabricated from recovered materials to the maximum extent practicable, provided the reeling machine produced meets all other requirements of this specification Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the reeling machine.

3.2 Lubrication. The wheel bearings shall be greased with grease conforming to MIL-G-10924.

3.3 Identification marking. The reeling machine shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514. plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.4 Treatment and painting. The portions of the reeling machine normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or type G as applicable, with the finish coat conforming to green 383.

3.5 Welding and welders.

3.5.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the reeling machine to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the reeling machine shall be at the option of the contractor.

3.5.1.1 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and carriage.

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3.5.1.2 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that., when the welded parts are assembled together to make the reeling machine, the reeling machine shall conform to the . inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.5.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. The certification shall be made available for review by the contracting officer or the contracting officer's representatives

3.6 Workmanship. All parts, components, and assemblies of the reeling machine including castings, forgings, molded parts, stamping, bearing seals, machined. surfaces, and welded parts shall be clean and freee from burrs, sharp edges, and corners, except when sharp edges and corners are required or where they are not detrimental to safety.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize-submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are. classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. Samples selected shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4 Examination procedure.

4.4.1 Examination. The reeling machine shall be examined as specified herein for the following defects:

MAJOR

101. Nonconformance to drawings.
102. Materials not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.1.2).
103. Dissimilar metals as specified in MIL-STD-889 not effectively insulated from each other as specified (see 3.1.3).
104. Contractor does not have documentation available for identification of material finishes, or treatments as specified (see 3.1.4).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the reeling machine (see 3.1.5).
106. Lubrication not as specified (see 3.2).
107. Welding not as specified (see 3.5).
108. Welder and welding certification not as specified (see 3.5.1 and 3.5.2).
109. Workmanship not as specified (see 3.6).

MINOR

201. Treatment and painting not as specified (see 3.4).
202. Identification marking missing, incomplete illegible, or not as specified (see 3.3).

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1. Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 1.0 defective.

110. Materials or containers not as specified for levels A and B (see 5.1.1.4 and 5.2).
111. Disassembly not as specified for level A (see 5.1.1.1).
112. Safety nuts not reinstalled on the axles for level A.
113. Unprotected metal surfaces not coated with applicable preservative for level A (see .5.1.1.3.).
114. Wheels, axles, and cable-reel shafts not preserved as specified for level A (see 5.1.1.4).
115. Openings of wheel hubs not covered for level A (see 5.1.l.4).
116. Strapping not as specified (see 5.2.1 and 5.2.2).
117. Marking illegible, incomplete, incorrect, or missing. (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.10 Level A.

5.1.1.1 Disassemble. The wheels shall be removed from the axle.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unprotected metal surfaces. Unprotected metal surfaces of the reeling machine requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative, except as specified herein.

5.1.1.4 Wheels, axles, and cable-reel shaft. The interior of the wheel hubs, the bearing surfaces of the axles, and the unpainted surfaces of the cable-reel shaft shall be coated with type P-6 preservative. The axle safety nuts shall be coated with type P-1 preservative and reinstalled on the axles. The coated surfaces of the axles and the cable-reel shaft shall be wrapped with barrier materials conforming to MIL-B-121, type I, grade A, class 2. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV. The hub openings shall be covered with barrier material specified herein. The barrier material

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shall be secured with tape conforming to PPP-T-60, type IV. Bearings, if removed, shall be coated with type P-6 preservative and preserved together in accordance with MIL-P-116, method IC-1. The preserved bearings shall be secured to the reeling machine in a protected location with tape conforming to PPP-T-60, type IV.

5.1.2 Commercial. Reeling machines shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The reeling machines shall be packed together in quantities not to exceed the weight limitation of close-fitting boxes conforming to PPP-B-601, Overseas type, style I or J. Blocking, bracing, anchoring, and cushioning shall be provided to prevent or control movement within the box. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The reeling machine shall be packed as specified in 5.2.1 for level A except the boxes shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. Reeling machines, preserved as specified in 5.1, shall be packed accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The reeling machine is intended for use in transporting cable reels to facilitate the laying and retrieving of electrical cable.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following.

a. Title, number, and date of this specification.
b. When the welding process is not at the option of the contractor (see 3.5.1).
c. Degree of preservation and degree of packing required (see 5.1 and 5.2).
d. When strapping shall be finish A (see 5.2.1).

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6.3 Subject term (key word) listing.

Cable
Hand
Reeling machine

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Army - AT

Preparing activity:
Army - ME

Project 3895-A236

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-52003D(ME)	Reeling Machine, Cable, Hand

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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